QPF Lab — Associator Spectroscopy (Final Engineering Spec)

Purpose. QPF is a falsifiable discovery procedure for perceptual composition. It learns (i) a minimal set of nonnegative perceptual “primes” and (ii) a lawful composition operator that predicts human judgments on mixtures. The *associator* is treated as a structural anomaly detector: persistent non-associativity is used to discover missing latent macrostates (e.g., context/adaptation), or else to certify irreducible non-associativity under a fixed class of lawful augmentations.

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1. Goals and falsifiable hypotheses

Goal

Discover a low-rank, interpretable percept basis and an admissible percept-composition law that predict out-of-sample responses to binary and ternary mixtures, while separating **stimulus mixing** from **percept mixing**.

Core hypotheses

- **H1 (Low-rank + lawful beats unconstrained):** A low-dimensional prime basis with a lawful composition operator predicts held-out judgments (especially ternary mixes) better than unconstrained baselines at matched complexity.
- **H2 (Netting selectivity):** Where cancellation is behaviorally licensed, extending the state to a group-complete form improves predictive evidence; otherwise it does not.
- **H3 (Closure defect):** Most measured non-associativity across triple mixes is a closure defect that collapses after adding a minimal latent macrostate (often context/adaptation).

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2. Mathematical foundation (Option A: latent-space order + extended cone)

2.1 Spaces

- **Stimulus space:** $S \subseteq \mathbb{R}^C_{\geq 0}$ (C stimulus channels).
- **Latent percept space:** $Z = \mathbb{R}^m_{\geq 0}$ with cone $K = Z$ and partial order $z \succeq z' \iff z - z' \in K$.

2.2 Prime dictionary (nonnegative factorization)

- Representation: $$ z = P a,\quad P \in \mathbb{R}^{m\times K}{\ge 0},\ a \in \mathbb{R}^K. $$}\ \text{(sparse)
- Practical parameterization: `P = softplus(P_raw)` and `a = softplus(a_raw)`.

Scale/identifiability anchors (required):

- Column-normalize dictionary (e.g., L1): $\sum_i P_{ik} = 1$ for each prime k.
- Normalize magnitude weights (e.g., $\|w\|_1 = 1$).

2.3 Magnitude and aspects

- **Magnitude functional:** $J(z) = w^\top z$, with $w \in \mathbb{R}^m_{\ge 0}$.
- **Aspect intensities:** $I_\ell(z) = u_\ell^\top z$, with $u_\ell \in \mathbb{R}^m_{\ge 0}$.

Immediate consequences:

- Monotonicity is structural if the operator preserves the cone order.
- Contractivity is meaningful and compatible with nonnegativity.

2.4 Netting-enabled aspects (extended cone)

For aspects where cancellation is behaviorally licensed, represent signed percepts as differences of positives:

$$ (z^+, z^-) \in K \times K,\quad z_{net} = z^+ - z^-. $$

Use magnitude on the extended state:

$$ J(z^+,z^-) = w^\top(z^+ + z^-) $$

so cancellation cannot be gamed via large opposing components.

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3. Percept composition operators (lawful by construction)

All operators take an explicit mix proportion $\alpha \in [0,1]$:

$$ \otimes_\alpha: Z\times Z \to Z. $$

3.1 LRS (Linear receptor-sum surrogate)

$$ \mathrm{LRS}_\alpha(z_1,z_2) = \alpha z_1 + (1-\alpha) z_2. $$

Guarantees: cone-preserving, order-preserving, and J-contractive.

3.2 DN (Divisive normalization) — two variants

Variant A: Coordinatewise DN (recommended for v1.0)

$$ \mathrm{DN}^{coord}_\alpha(z_1,z_2) = \frac{n}{1+\sigma + \gamma n},\quad n=\alpha z_1+(1-\alpha)z_2,\ \sigma,\gamma\ge 0. $$

Guarantees: cone-preserving, order-preserving, and J-contractive.

Variant B: Global DN (optional menu item)

$$ \mathrm{DN}^{global}_\alpha(z_1,z_2) = \frac{n}{1+\sigma + \gamma\,J(n)},\quad n=\alpha z_1+(1-\alpha)z_2. $$

Guarantees: cone-preserving and J-contractive due to denominator ≥ 1. **Not guaranteed order-preserving** under the cone order; treat monotonicity as a diagnostic/penalty when using this operator.

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4. Context / adaptation macrostate (Patch A: fixed dimensionality)

4.1 Scalar context state

Project stimulus to a nonnegative scalar signal:

$$ \xi(s) = q^\top s,\quad q\ge 0. $$

Exponential smoothing update:

$$ h_{t+1} = \lambda h_t + (1-\lambda)\,\xi(s_t),\quad \lambda\in[0,1]. $$

4.2 Context injection (cone-lawful)

- **Additive:** $z_t = Pa_t + vh_t$, with $v \ge 0$.
- **Divisive (often more realistic):** $z_t = (Pa_t)/(1+\eta h_t)$, with $\eta \ge 0$.

Note on suppressive adaptation

If adaptation must subtract directionally, implement it in the **extended cone** (signed) state, or as multiplicative gain control (divisive), not as subtraction in Z.

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5. Carrier invariance (correct invariants)

A strict identity element for all $\alpha \neq 1$ is generally incompatible with convex-mixture operators. QPF v1.0 uses:

1. **Endpoint identity:**

2. $\otimes_{\alpha=1}(z_1, z_2) = z_1$

3. $\otimes_{\alpha=0}(z_1, z_2) = z_2$

4. **Carrier “no-new-direction” invariance:** Let the carrier encode to $z_0 \approx 0$. Then

$$ \otimes_\alpha(z,z_0) \approx \kappa(\alpha,z)\,z,\quad \kappa\ge 0. $$

Implementation check: high cosine similarity between normalized z and $\otimes_\alpha(z, z_0)$.

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6. Associator spectroscopy (discovery loop)

6.1 Associator definition

For triples (z_1, z_2, z_3) and mix proportions (α, β):

$$ A = d\Big(R(\otimes_\beta(\otimes_\alpha(z_1,z_2),z_3)),\ R(\otimes_\alpha(z_1,\otimes_\beta(z_2,z_3))) \Big) $$

- **Observable distance** d (v1.0): L2 difference in predicted aspect means plus triangle-test cross-entropy.

6.2 Noise-calibrated signal

Estimate associator noise floor $\widehat{\sigma}_A$ via repeats / bootstrap.

$$ S(M) = \frac{\mathbb{E}[A]}{\widehat{\sigma}_A} $$

Stop when $S(M) \le \tau$ (e.g., $\tau = 1.1$).

6.3 Minimal augmentation menu (cone-lawful)

Each augmentation must preserve cone structure by construction.

- `context_1d`: add h with positive projection q and injection v (additive or divisive).
- `extra_prime`: increase $K \to K + 1$ (nonnegative new column and activation).
- `subject_slope`: subject-specific positive scaling on u_ℓ or on readout gains.
- `operator_mixing`: convex mix of LRS and DN, $\beta \in [0, 1]$.
- `sparse_interaction`: safe, bounded, nonnegative interaction term (typically with additional normalization to avoid blow-up).

6.4 Selection criterion (“no free wins”)

Adopt an augmentation only if:

- held-out **ternary** ELPD improves (veto otherwise),
- associator signal S decreases meaningfully,
- complexity-adjusted score improves:

$$ \Delta\mathrm{Score} = \Delta\mathrm{ELPD}_{ternary} - \lambda\,\Delta\mathrm{Complexity} - \gamma\,\Delta S $$

If no augmentation improves score, declare **irreducible non-associativity** under the menu (scientific result).

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7. Training objective (v1.0)

Let data include ratings y and triangle outcomes t.

Likelihood:

- Ratings: Gaussian or ordinal-logit (depending on scale).
- Triangle test: multinomial / logistic.

Regularization / priors (implemented as penalties):

- Sparsity on a: L1 or structured sparse gates.
- Carrier-null penalty: encourage $z_{carrier} \approx 0$.
- Optional continuity penalty: Lipschitz surrogate on readouts.
- Monotonicity penalty: only required if using `DN_global` or other non-monotone operators.

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8. Unit-test invariants (run during training)

Hard asserts for `LRS` and `DN_coord`; diagnostics (soft checks) for `DN_global`.

1. **Cone closure:** $\otimes_\alpha(z_1, z_2) \in K$.
2. **Endpoint identity:** $\otimes_1(z_1, z_2) = z_1$, $\otimes_0(z_1, z_2) = z_2$.
3. **J-contractivity:** $J(\otimes_\alpha(z_1, z_2)) \le \max(J(z_1), J(z_2))$.
4. **Carrier no-new-direction:** cosine similarity between normalized z and $\otimes_\alpha(z, z_0)$ near 1.
5. **Order preservation (when guaranteed):** if $z_1' \succeq z_1$, then $I(\otimes_\alpha(z_1', z_2)) \ge I(\otimes_\alpha(z_1, z_2))$.

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9. Implementation plan (week 1–2)

Week 1

- Implement `QPFBase`: softplus constraints, column normalization for `P`, normalized `w`, positive `U`.
- Implement `LRS` and `DN_coord` (plus optional `DN_global`).
- Implement `associator()` and noise-floor bootstrap.
- Add carrier-null penalty and L1 sparsity on `a`.
- Add invariant tests to CI / training loop.

Week 2

- Implement `context_1d` augmentation wrapper (projection q, state h, injection).
- Implement scoring + veto rule (ternary ELPD must not degrade).
- Add associator heatmaps and "associator spectroscopy" iteration driver.

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10. Minimal dataset schema (starter)

- **subjects:** `subject_id`, demographics, session info
- **stimuli:** `stimulus_id`, `component_vector` (length C), `intensity_level`, `carrier_id`
- **trials:** `trial_id`, `subject_id`, `task_type` {rating, triangle, similarity}, `(s_a_id,s_b_id,alpha)` and optional `(s_c_id,beta)`; presentation order; timestamp
- **responses:** ratings per aspect; triangle choice/correct; similarity score/choice
- **aspects:** `aspect_id`, scale spec, `netting_allowed` flag

External references (selected)

Divisive normalization / canonical computations

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